Machine Learning – CDS201

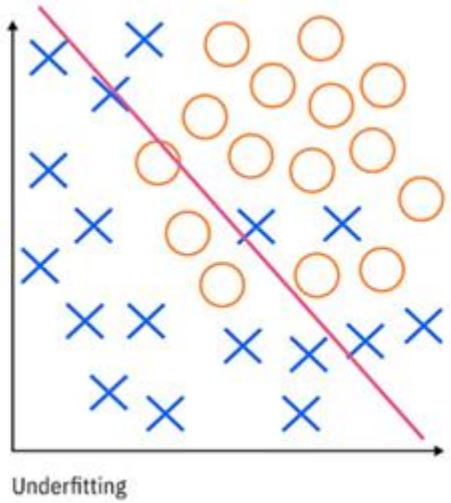
Module 2: Regression Models

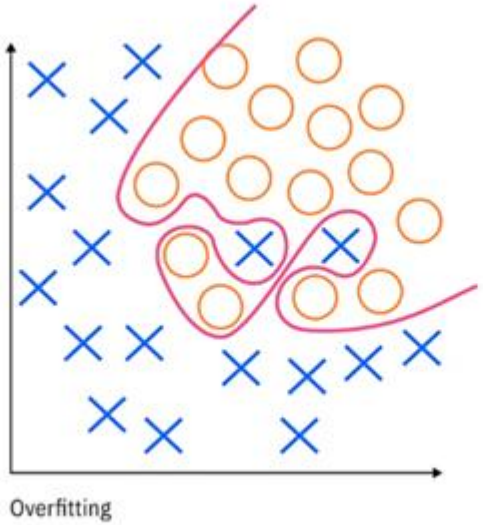
Machine learning Terminologies

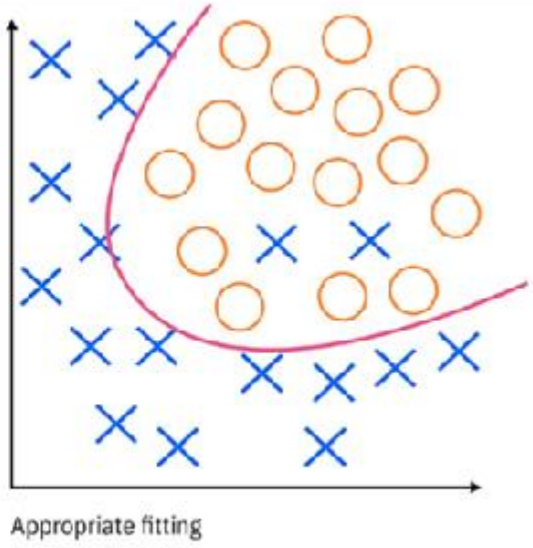
- Generalization
- Overfitting
- Underfitting

- Bias & Variance

- Overfitting: Training error is low, but testing error is significantly higher.
- Underfitting: Errors are consistently high across training and testing data sets.
- Generalization is the model's ability to understand and apply learned patterns to unseen data.

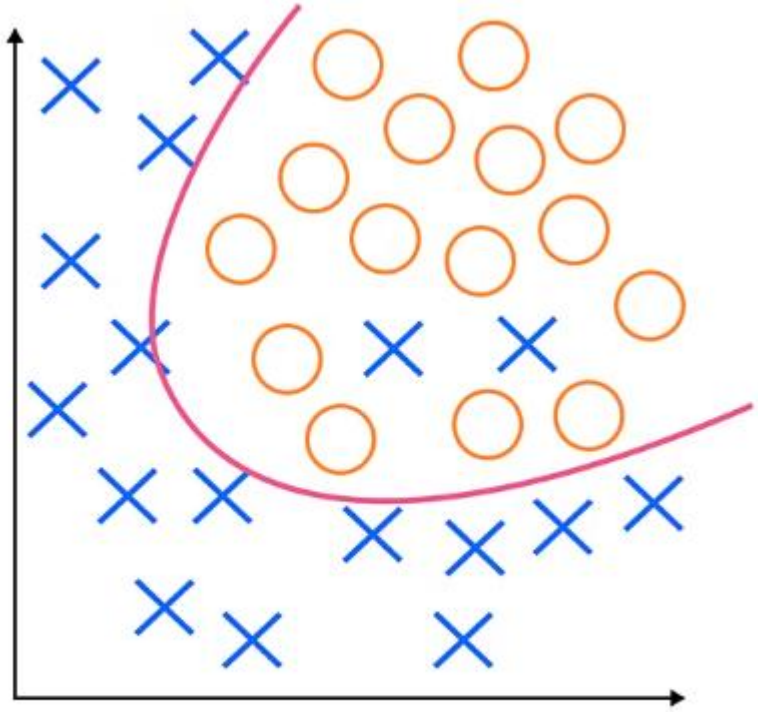




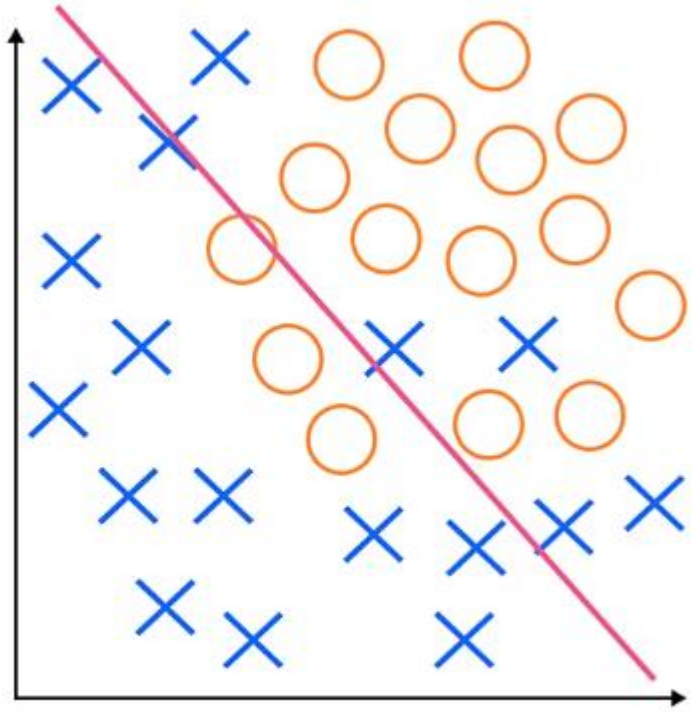


- An **overfit model** can result in high model accuracy on training data but low accuracy on new data due to memorization instead of generalization.
- When trained on a small or noisy data set, the model risks memorizing specific data points and noise rather than learning the general patterns.
- Another sign of an overfit model is its decision boundaries, the model's learned rules for classifying data points. The decision boundary becomes overly complex and erratic in overfit models, as it adapts to noise in the training set rather than capturing true underlying structures, further indicating overfitting.
- The model exhibits **low bias and high variance.**

- An **underfit model** performs poorly on training data and testing data because it fails to capture the dominant patterns in the data set.
- Underfit models also tend to show high errors in learning curves, return suboptimal evaluation metrics and exhibit systematic residual patterns, all of which indicate an inability to learn the underlying relationships in the data effectively.
- Underfitting in machine learning often occurs due to simplistic models, poor feature engineering or excessive regularization that overly restricts the model's flexibility. Similarly, poor feature selection—such as omitting interaction terms or polynomial features—can prevent the model from understanding hidden relationships in the data. Inadequate preprocessing, insufficient training time or a lack of sufficient data to train the model can also contribute to underfitting.
- The model exhibits **high bias and high variance.**



Appropriate fitting



Underfitting

Bias and variance

- A biased model makes strong assumptions about the training data to simplify the learning process, ignoring complexities it cannot account for.
- Variance refers to the model's sensitivity to learning fluctuations in the training data (which is actually your test data).
- Models with low variance also tend to underfit as they are too simple to capture complex patterns. However, low-bias models might overfit if they are too flexible.

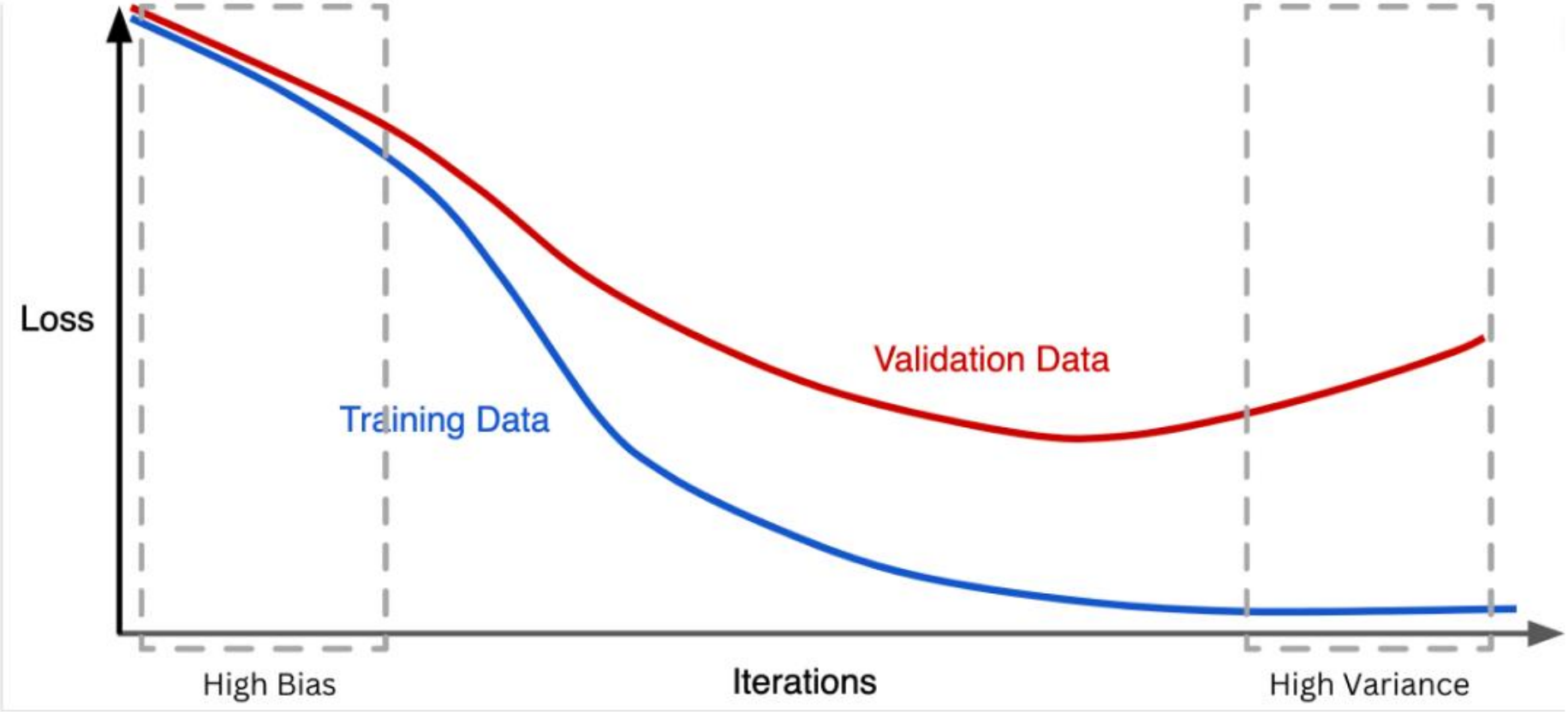
Bias and variance

- High variance indicates that the model might capture noise, idiosyncrasies and random details within the training data.
- High-variance models are overly flexible, resulting in low training error, but when tested on new data, the learned patterns fail to generalize, leading to high test error.
- Imagine memorizing answers for a test instead of understanding the concepts needed to get the answers yourself. If the test differs from what was studied, you will struggle to answer the questions.
- Striking the balance between variance and bias is key to achieving optimal performance in machine learning models.

Bias and variance

- High Bias, Low Variance: The models are consistent but inaccurate on average, typical of overly simplified models.
- Low Bias, High Variance: Models are accurate on average but inconsistent across different datasets. This is typical of overly complex models.
- Low Bias, Low Variance: Models are accurate and consistent on training and new data, indicating a good balance between model complexity and performance on unseen data.
- High Bias, High Variance: Models are inaccurate and inconsistent, performing poorly in training and on new data.

- Underfitting: Occurs when the model is too simple, characterized by low variance and high bias.
- Overfitting: Occurs when the model is too complex, characterized by high variance and low bias.


Loss
Validation Data
Training Data
High Bias
Iterations
High Variance

Strategies to Avoid Underfitting and Overfitting

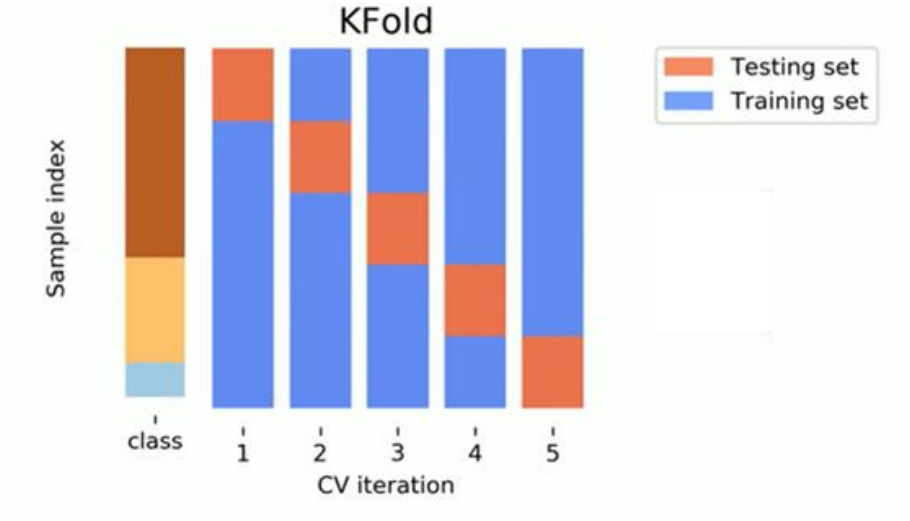
- Use Cross-Validation
- Regularization
- Feature Selection
- Increase Training Data
- Early Stopping

Cross validation

- Cross-validation determines the accuracy of your machine learning model by separating the data into two groups: the training set and the testing set.
- Cross-validation can help you address problems such as overfitting or underfitting a model.
- Provides a more accurate estimate of your model's ability to perform on new or unseen data by providing you with an unbiased estimate of the generalization error (a measure of how well an algorithm predicts future observations for previously unseen data).
- It involves partitioning the original dataset into multiple subsets (called *folds*), training the model on a few of them, and validating it on the remaining ones.

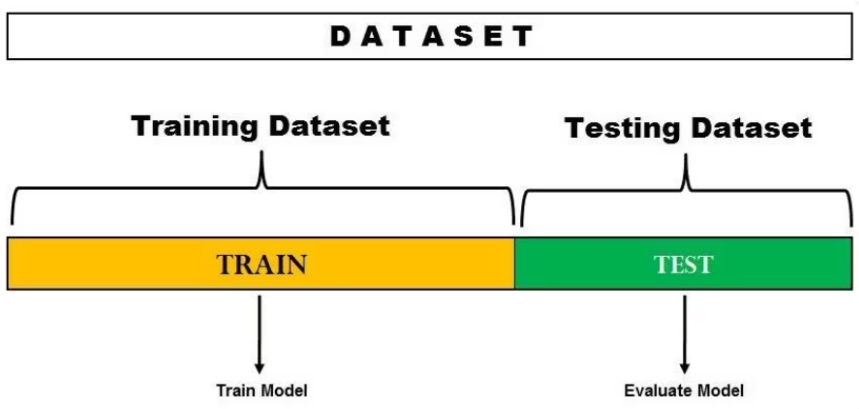
K-fold cross-validation

- The whole dataset is partitioned in k parts of equal size and each partition is called a fold. It's known as k-fold since there are k parts where k can be any integer - 3,4,5, etc.
- Small k (e.g., 3 or 5): High bias, low variance
- Large k (e.g., 10): Low bias, higher variance
- k ≈ 10 is often a sweet spot, offering a good balance.



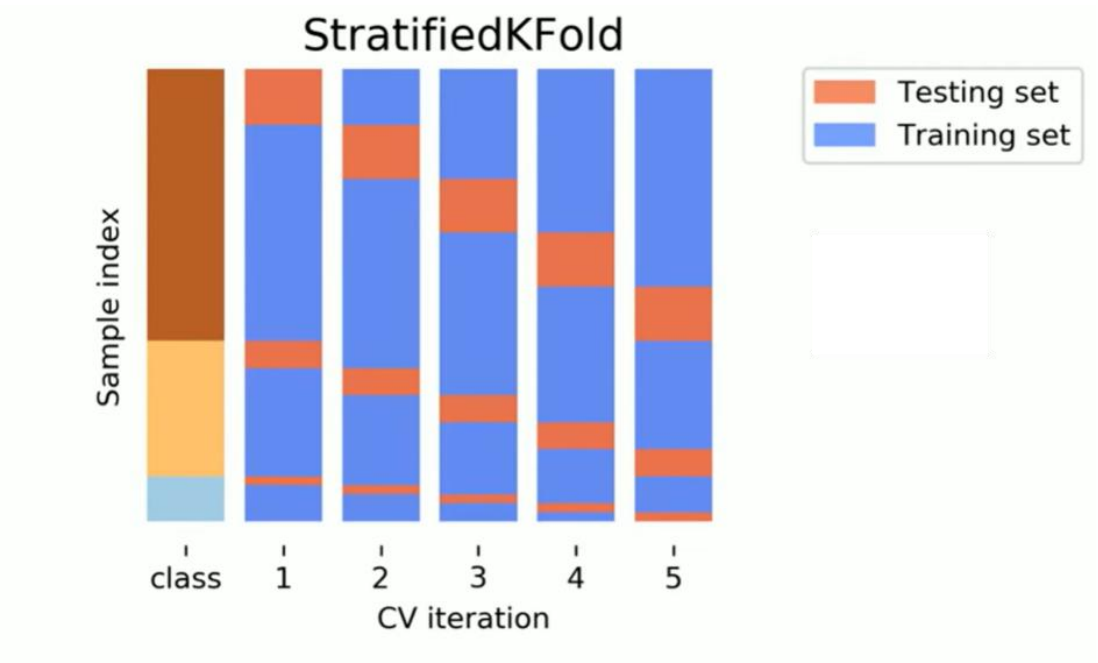
Holdout cross-validation

- Holdout cross-validation has the entire dataset partitioned randomly into a training set and a validation set.
- A rule of thumb to partition data is that nearly 70% of the whole dataset will be used as a training set and the remaining 30% will be used as a validation set.
- Since the dataset is split into only two sets, the model is built just one time on the training set and executed faster.



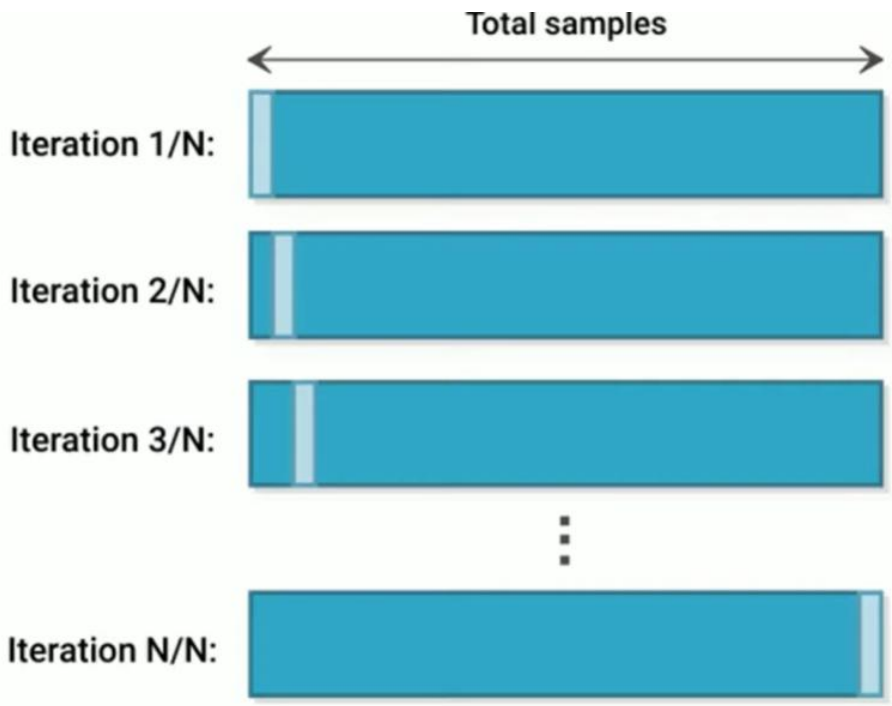
Stratified K-Fold Cross Validation

- Ensures that each fold maintains the same class distribution as the original dataset.
- **Ideal for classification problems**, especially with imbalanced classes



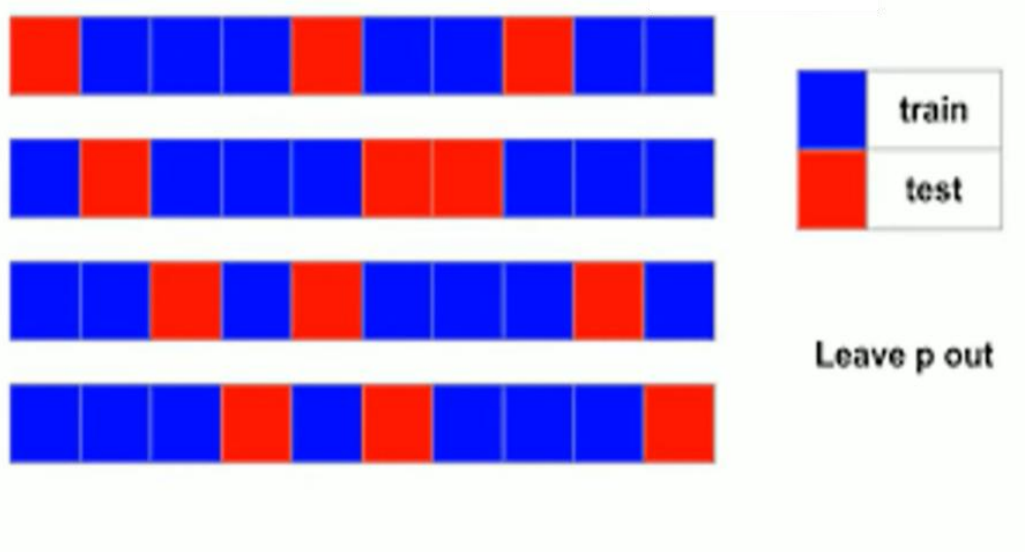
Leave-One-Out Cross Validation (LOOCV)

- Trains the model on n−1 data points, leaving out one as the test set.
- This is repeated n times (once for each data point).
- **Pros:** Utilizes the entire dataset, minimal bias.
- **Cons:** Extremely slow for large datasets; high variance due to small test set size.



Leave-P-Out Cross Validation

- Similar to LOOCV, but instead of 1, it leaves out *p* data points.
- This continues till all samples are used as a test set.
- However, it's not considered ideal for an imbalanced dataset and is deemed to be a computationally unfeasible method. This is because if the training set has all samples of one class, the model will not be able to properly generalize and will become biased to either of the classes.



When to Use What?

- LOOCV / Leave-p-out: Only for very small datasets where computation isn't an issue.
- K-Fold: Most practical and widely used. Stick to k=10 for balanced results.
- Stratified K-Fold: Must-use for classification problems with imbalanced data.

Method	Bias	Variance	Computational Cost	Best For
LOOCV	Low	High	Very High	Small datasets
Leave-p-Out	Low	High	Extremely High	Research, very small datasets
K-Fold (k = 3 or 5)	High	Low	Moderate	General use
K-Fold (k ≈ 10)	Balanced	Balanced	Moderate	Most problems
Stratified K-Fold	Balanced	Balanced	Moderate	Classification, class imbalance